GOOGLE | 1POINT3ACRES | TOPIC 4 OF 10 | CORE TOPIC

# Heaps, Stacks and Queues

11 questions to study here, each on its own page, with the full interview answer and code

3 more fit this topic but are covered in another topic: they are listed as SKIP at the end of the contents

| 1 | 10 |
|---|---|
| HARD | MEDIUM |

**How to use**

- Questions are Hard first, then Medium, then Easy, newest last-asked date first.
- PDFs run from the one with the most questions to the fewest. A question has pages in only one PDF; the others list it as SKIP
- Every page has < Prev | Contents | Next > at the bottom. Tap a title in the Contents to jump to it.
- Select code and press Cmd/Ctrl+C to copy it.

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# Contents

**Heaps, Stacks and Queues | 11 questions | tap a title to jump to it**


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**STUDY HERE: topic 4, Heaps, Stacks and Queues**

Also fits: Binary Search and Sorting (listed there as SKIP, no pages). Site tags: binary-search, heap, divide-and-conquer, hard

HARD | IN APP LC 4 | 1Point3Acres | tech-screen | 1 of 11

# Median of Two Sorted Arrays (LC 4) with a Follow-up Variant

ASKED 2026-10-09 | 11 DAYS AGO | LAST 30 DAYS

STUDY: TYPE | ALL TIME 74.2 | 30 DAYS 61 | PRIORITY 4/228

## ABOUT THIS QUESTION

**Topic:** Heaps, Stacks and Queues

**Optimal answer from:** your app's interview approach (LeetCode #4)

**LeetCode match:** LC 4 Median of Two Sorted Arrays

Frequency: Single report | Last asked: 2026-10-09 | Roles: SWE

**Original:** https://www.1point3acres.com/interview/problems/company/google/median-two-sorted-arrays-k-inputs

## QUESTION

L5 R1 coding: LC 4 (Median of Two Sorted Arrays). Heading toward the LC 295 streaming two-heap approach drew a redirect from the interviewer. Follow-up: extend to many sorted inputs, in the spirit of LC 23 (Merge k Sorted Lists).

Per the candidate, the base question is LC 4 (Median of Two Sorted Arrays): given two sorted arrays, return the median of their combined elements.
**Follow-up:** the input becomes many sorted arrays instead of two, in the spirit of LC 23 (Merge k Sorted Lists).

## APPROACH

Drill LC 4 until you can explain the partition invariant and the odd/even median cases without notes.
Drill LC 23 with a min-heap and with divide-and-conquer merging, and compare their costs as the number of inputs grows.
Practice stating, in under a minute, why a streaming two-heap approach is unnecessary when the inputs are static and sorted.

## INTERVIEW-DAY PLAN

### CLARIFY FIRST

Ask for the median definition when the total is even (the mean of the two middle values), whether the arrays can be empty, whether duplicates are possible, and the sizes.

### BRUTE FORCE TO SAY OUT LOUD

Merge and sort, then read the middle. O((m + n) log(m + n)); merging two sorted arrays with two pointers is O(m + n).

### OPTIMAL IDEA

Binary search the cut in the shorter array: choose i in a and j = (m + n + 1) // 2 - i in b so the left parts hold half of the elements. The cut is valid when a[i-1] <= b[j] and b[j-1] <= a[i], with minus and plus infinity at the ends. The answer is the larger left value for an odd total, or the mean of the larger left and the smaller right value. O(log min(m, n)). For k arrays: binary search on the value and count elements at most x with a bisect per array, O(k log n log range), or pop from a min-heap until the middle, O(N log k).

### DRY-RUN THESE CASES

One array empty, both of equal length, total odd and even, all elements equal, one array entirely smaller than the other, negative numbers.

### LIKELY FOLLOW-UPS

Many sorted arrays (the stated follow-up), a stream (two heaps), or the k-th element instead of the median.

### PITFALLS

Starting from the two-heap design of the streaming median (wrong family for static sorted input). Searching the longer array and running past its end. Off-by-one in the half size for odd totals.

## HOW TO THINK ABOUT IT

Both inputs are already sorted, so the median is a partition problem, not a streaming one (the two-heap design of LeetCode 295 is the wrong family and the interviewer redirects it). Split the small array at i and the large one at j so that the left halves together hold half the elements; the split is right when every left value is at most every right value, which only needs two comparisons across the cut. That condition is monotone in i, so binary search it in O(log min(m, n)). For k arrays the same idea becomes: binary search on the value and count elements at most x with one bisect per array.

**Basis:** LeetCode 4, Median of Two Sorted Arrays; the follow-up to many sorted inputs is in the spirit of LeetCode 23, Merge k Sorted Lists.

## BRUTE FORCE

### BRUTE FORCE · PSEUDO-CODE

```
def median(a, b):
    merged = sorted(a + b)
    n, mid = len(merged), len(merged) // 2
    if n % 2:
        return merged[mid]
    return (merged[mid - 1] + merged[mid]) / 2
```

| TIME | SPACE |
|---|---|
| O((m + n) log(m + n)) | O(m + n) |

### WHY IT IS SLOW

Sorting ignores that the inputs are sorted and costs O((m + n) log(m + n)) time and O(m + n) memory. A binary search for the partition reads only O(log min(m, n)) values.

## OPTIMAL SOLUTION · Binary search on the partition

| TIME | SPACE |
|---|---|
| O(log min(m, n)) | O(1) |

**Code:** see FULL INTERVIEW CODE below (the optimal solution is its main function).

## FOLLOW-UP 1 Now there are k sorted arrays instead of two. How do you find the median?

Binary search on the answer value. For a candidate x, count the elements at most x with one bisect per array (O(k log n)); the median is the smallest x whose count reaches the middle rank. With integers the search range is the global min to max, so the cost is O(k log n log(range)). The heap alternative pops the smallest element from a k-way merge until it passes the middle, O((N / 2) log k), which is simpler but linear in the total size.

## FOLLOW-UP 2 What is the heap version and when would you choose it?

A min-heap holds the front of every array. Pop (N + 1) / 2 times, remembering the one or two middle values. It is O(N log k) at worst, trivial to get right and needs no bound on the values, so choose it when the arrays are small or the values are not integers; choose the value search when N is huge and k is small.

## FULL INTERVIEW CODE · base question + every follow-up

**One program for the whole question:** the base answer and every follow-up above, built from small helper functions, with a comment above each step. Type it top to bottom in the interview; the demo at the bottom runs it.

```
import bisect
import heapq


# BRUTE FORCE (say it first, it also checks the answers)
def median_brute(*arrays):
    # the oracle: merge, sort, read the middle
    merged = sorted(x for a in arrays for x in a)
    n, mid = len(merged), len(merged) // 2
    if n % 2:
        return merged[mid]
    return (merged[mid - 1] + merged[mid]) / 2


# HELPERS for the base question
INF = float("inf")


# BASE QUESTION (optimal): Binary search on the partition
def median_two(a, b):
    """Both arrays are sorted: binary search the cut. The
    left parts must hold half of all the elements."""
    if len(a) > len(b):
        a, b = b, a
    m, n = len(a), len(b)
    if m + n == 0:
        raise ValueError("no elements")
    lo, hi, half = 0, m, (m + n + 1) // 2
    while lo <= hi:
        i = (lo + hi) // 2
        j = half - i
        # values just left and right of each cut
        a_left = a[i - 1] if i > 0 else -INF
        a_right = a[i] if i < m else INF
        b_left = b[j - 1] if j > 0 else -INF
        b_right = b[j] if j < n else INF
        if a_left <= b_right and b_left <= a_right:
            # a valid cut: read the median off its edges
            if (m + n) % 2:
                return max(a_left, b_left)
            return (max(a_left, b_left)
                    + min(a_right, b_right)) / 2
        if a_left > b_right:
            hi = i - 1
        else:
            lo = i + 1


# FOLLOW-UP 1: Now there are k sorted arrays instead of
# two. How do you find the ...
def median_k_value(arrays):
    """k arrays: binary search the value; one bisect per
    array counts how many elements are at most x."""
    arrays = [a for a in arrays if a]
    total = sum(len(a) for a in arrays)
    if total == 0:
        raise ValueError("no elements")

    def kth(rank):
        # smallest x with at least `rank` elements <= x
        lo = min(a[0] for a in arrays)
        hi = max(a[-1] for a in arrays)
        while lo < hi:
            mid = (lo + hi) // 2
            count = sum(bisect.bisect_right(a, mid)
                        for a in arrays)
            if count >= rank:
                hi = mid
            else:
                lo = mid + 1
        return lo

    if total % 2:
        return kth(total // 2 + 1)
    return (kth(total // 2) + kth(total // 2 + 1)) / 2


# FOLLOW-UP 2: What is the heap version and when would you
# choose it?
def median_k_heap(arrays):
    """k arrays, simple version: pop from a k-way merge
    until the middle one or two values have been seen."""
    heap = [(a[0], i, 0)
            for i, a in enumerate(arrays) if a]
    heapq.heapify(heap)
    total = sum(len(a) for a in arrays)
    if total == 0:
        raise ValueError("no elements")
    low_rank, high_rank = (total - 1) // 2, total // 2
    seen, middle = 0, []
    while heap:
        value, i, j = heapq.heappop(heap)
        if seen >= low_rank:
            middle.append(value)
        if seen >= high_rank:
            break
        seen += 1
        if j + 1 < len(arrays[i]):
```

```
            nxt = (arrays[i][j + 1], i, j + 1)
            heapq.heappush(heap, nxt)
    if total % 2:
        return middle[0]
    return (middle[0] + middle[1]) / 2


if __name__ == "__main__":
    assert median_two([1, 3], [2]) == 2
    assert median_two([1, 2], [3, 4]) == 2.5
    assert median_two([], [5]) == 5
    assert median_k_value([[1, 3], [2], [4, 5, 6]]) == 3.5
    assert median_k_heap([[1, 3], [2], [4, 5, 6]]) == 3.5
    import random
    for _ in range(300):
        arrays = []
        for _ in range(random.randint(1, 5)):
            size = random.randint(0, 6)
            arrays.append(sorted(random.randint(-9, 9)
                                 for _ in range(size)))
        if not any(arrays):
            continue
        want = median_brute(*arrays)
        assert median_k_value(arrays) == want
        assert median_k_heap(arrays) == want
        if len(arrays) >= 2:
            rest = sum(arrays[1:], [])
            rest.sort()
            assert median_two(arrays[0], rest) == want
```

## NOTES FROM CANDIDATES

Pitfalls: Starting from the LC 295 (Find Median from Data Stream) two-heap design is the wrong family for the base question; the interviewer stepped in with a hint. Both inputs are already sorted — lean on that.

Follow-ups: Many sorted inputs instead of two

## PREPARATION

Drill LC 4 until you can explain the partition invariant and the odd/even median cases without notes.
Drill LC 23 with a min-heap and with divide-and-conquer merging, and compare their costs as the number of inputs grows.
Practice stating, in under a minute, why a streaming two-heap approach is unnecessary when the inputs are static and sorted.

**STUDY HERE: topic 4, Heaps, Stacks and Queues**

Also fits: Arrays, Hashing and Two Pointers (listed there as SKIP, no pages). Site tags: array, monotonic-stack, stack, medium

MEDIUM | 1Point3Acres | tech-screen | 2 of 11

# Visible People in a Line (Last Person, Then Everyone)

ASKED 2026-10-07 | 13 DAYS AGO | LAST 30 DAYS

STUDY: TALK | NO LEETCODE FREQUENCY | PRIORITY 181/228

## ABOUT THIS QUESTION

**Topic:** Heaps, Stacks and Queues

**Optimal answer from:** written and tested here for this exact question

**Related in your app (a neighbouring problem, not this exact question):** LeetCode #1944 Number of Visible People in a Queue

**LeetCode match:** LC 1944 Number of Visible People in a Queue (related)

Frequency: Single report | Last asked: 2026-10-07 | Roles: SWE

**Original:** https://www.1point3acres.com/interview/problems/company/google/visible-people-in-line

## QUESTION

Given heights of people in a line, A can see B if nobody between them is taller than both. Part 1: count how many people the last person can see. Part 2: return that count for every person. Written in a plain editor with no compilation.

**Input:** an array of heights representing a line of people.
Visibility rule: person A can see person B if no one standing between them is taller than both A and B.
**Part 1:** return how many people the last person in line can see.
**Part 2:** for the same array, return, for every person in the line, how many people they can see.

### EXAMPLES

```
[1, 10, 6, 7, 9, 8, 2, 4, 3, 5] -> 6 (the last person,
    height 5, can see 3, 4, 2, 8, 9, 10).
```

## APPROACH

Write a brute-force Part 1 that checks every earlier person against the rule, then look for a single-pass version before attempting Part 2.
Extend the Part 1 solution to Part 2 and state its complexity before coding.
Hand-trace the example above without running code, including a tie case you define yourself.

## INTERVIEW-DAY PLAN

### CLARIFY FIRST

Ask what blocks the view (someone strictly taller than both), how ties are treated, which directions count in Part 2, and whether the code runs (it does not: trace by hand).

### BRUTE FORCE TO SAY OUT LOUD

Check every pair against the people between them. O(n^2) for the last person, O(n^3) for everyone.

### OPTIMAL IDEA

Part 1: walk from the person before the last towards the front, keeping the maximum height between the last person and the current one; the current person is visible when that maximum is at most the larger of the two heights. O(n). Part 2: for each person, everyone up to the first strictly taller person on the right is visible, then only people at least as tall as everyone from there on; a monotonic stack gives the next at-least-as-tall and next strictly-taller indexes and the chain length in O(n). Run it on the reversed line for the left side and add.

### DRY-RUN THESE CASES

A single person, all equal heights, strictly increasing and decreasing lines, the example from the statement (the last person sees 6), a tie case you invent.

### LIKELY FOLLOW-UPS

Only looking in one direction, blocking on equal heights, or returning who they see instead of how many.

### PITFALLS

Using a strict comparison where the rule says taller than both. Forgetting that the first taller person is itself visible. Rereading the example without tracing it by hand.

## HOW TO THINK ABOUT IT

Person A sees person B when nobody strictly between them is taller than both, which means the tallest person between them is at most as tall as the taller of A and B. Walking away from the last person and keeping the running maximum of everyone passed lets you test each person in O(1). For everyone at once the visible people to the right form a chain: everything up to the first taller person, then only the people who are at least as tall as everyone before them. A monotonic stack builds those chains for all positions in O(n).

**Basis:** LeetCode 1944, Number of Visible People in a Queue, with the rule that a person is hidden only by someone taller than both.

## BRUTE FORCE

### BRUTE FORCE · PSEUDO-CODE

```
def sees(a, i, j):
    lo, hi = min(i, j), max(i, j)
    for k in range(lo + 1, hi):
        if a[k] > max(a[i], a[j]):
            return False
    return True
def count_for(a, i):
    return sum(1 for j in range(len(a))
               if j != i and sees(a, i, j))
```

| TIME | SPACE |
|---|---|
| O(n^3) for everyone, O(n^2) for the last person | O(1) |

## BRUTE FORCE

WHY IT IS SLOW

Each pair rescans the people between it, O(n^3) for everyone and O(n^2) for the last person. One pass with a running maximum answers the last person in O(n), and a monotonic stack answers everyone in O(n).

## OPTIMAL SOLUTION · One pass with a running maximum (last person)

| TIME | SPACE |
|---|---|
| O(n) | O(1) |

**Code:** see FULL INTERVIEW CODE below (the optimal solution is its main function).

## FOLLOW-UP 1 Part 2: return the number of people each person can see.

Run the right-looking count for every position with a monotonic stack, run it again on the reversed line for the looking-left count, and add the two. The stack makes both passes O(n), instead of the O(n^2) you would get by repeating the last-person scan for every position.

## FOLLOW-UP 2 How do ties and the direction of sight change the answer?

Under the stated rule only someone strictly taller than both blocks the view, so equal heights never block: in [5, 5, 5] the first person sees both others. If the interviewer wants 'taller or equal blocks', replace the non-strict comparisons with strict ones. If a person only looks towards the front, drop the reversed pass and use see_right on the reversed line alone.

## FULL INTERVIEW CODE · base question + every follow-up

**One program for the whole question:** the base answer and every follow-up above, built from small helper functions, with a comment above each step. Type it top to bottom in the interview; the demo at the bottom runs it.

```
# BRUTE FORCE (say it first, it also checks the answers)
def sees(a, i, j):
    # the oracle: nobody strictly between is taller than
    # both
    lo, hi = min(i, j), max(i, j)
    return all(a[k] <= max(a[i], a[j])
               for k in range(lo + 1, hi))


# BRUTE FORCE (say it first, it also checks the answers)
def counts_brute(a):
    return [sum(1 for j in range(len(a))
                if j != i and sees(a, i, j))
            for i in range(len(a))]


# HELPERS for the base question
INF = float("inf")


# BASE QUESTION (optimal): One pass with a running maximum
# (last person)
def last_sees(a):
    """Part 1: walk away from the last person keeping the
    tallest height seen strictly between us and them."""
    if not a:
        return 0
    last, between, count = a[-1], -INF, 0
    for k in range(len(a) - 2, -1, -1):
        # nobody between is taller than both of us
        if between <= max(last, a[k]):
            count += 1
        between = max(between, a[k])
    return count


# FOLLOW-UP 1: Part 2: return the number of people each
# person can see.
def see_right(a):
    """Part 2, looking right: everyone up to the first
    strictly taller person is visible, then only people at
    least as tall as everyone from that person on."""
    n = len(a)
    next_ge, next_gt = [None] * n, [None] * n
    stack = []
    for k in range(n - 1, -1, -1):
        # next person at least as tall as person k
        while stack and a[stack[-1]] < a[k]:
            stack.pop()
        next_ge[k] = stack[-1] if stack else None
        stack.append(k)
    stack = []
    for k in range(n - 1, -1, -1):
        # next person strictly taller than person k
        while stack and a[stack[-1]] <= a[k]:
            stack.pop()
        next_gt[k] = stack[-1] if stack else None
        stack.append(k)
    chain = [0] * n
    for k in range(n - 1, -1, -1):
        # how many "at least as tall" steps follow k
        if next_ge[k] is not None:
            chain[k] = 1 + chain[next_ge[k]]
    out = []
    for i in range(n):
        t = next_gt[i]
        if t is None:
            out.append(n - 1 - i)
        else:
            out.append(t - i + chain[t])
    return out


# FOLLOW-UP 1: Part 2: return the number of people each
# person can see.
def see_all(a):
    # looking left is looking right on the reversed line
    left = see_right(a[::-1])[::-1]
    return [x + y for x, y in zip(left, see_right(a))]


if __name__ == "__main__":
    line = [1, 10, 6, 7, 9, 8, 2, 4, 3, 5]
    assert last_sees(line) == 6
    assert see_all(line)[-1] == 6
    assert last_sees([]) == 0 and last_sees([7]) == 0
    # equal heights never block each other
    assert last_sees([5, 5, 5]) == 2
    assert see_all([5, 5, 5]) == [2, 2, 2]
    import random
```

```
    for _ in range(300):
        a = [random.randint(1, 6)
             for _ in range(random.randint(0, 9))]
        assert see_all(a) == counts_brute(a)
        if a:
            assert last_sees(a) == counts_brute(a)[-1]
```

## NOTES FROM CANDIDATES

Pitfalls: The code is written in a plain text editor that does not compile or run; trace the example by hand before declaring the solution done.
Confirm the direction of "see" for Part 2 (toward the front only, or both directions) and how equal heights are treated before coding; the example does not exercise ties.

Follow-ups: Generalize from the last person to every person in the line (Part 2).

## PREPARATION

Write a brute-force Part 1 that checks every earlier person against the rule, then look for a single-pass version before attempting Part 2.
Extend the Part 1 solution to Part 2 and state its complexity before coding.
Hand-trace the example above without running code, including a tie case you define yourself.

**STUDY HERE: topic 4, Heaps, Stacks and Queues**

Also fits: Arrays, Hashing and Two Pointers, Greedy and Math (listed there as SKIP, no pages). Site tags: interval, heap, greedy, medium

**MEDIUM | IN APP LC 253 | 1Point3Acres | phone-screen, tech-screen, onsite-coding | 3 of 11**

# Meeting Rooms / Interval Scheduling Variants

**ASKED 2026-09-29 | 21 DAYS AGO | LAST 30 DAYS**

**STUDY: TYPE** **ALL TIME 58** **30 DAYS 0** **PRIORITY 44/228**

## ABOUT THIS QUESTION

**Topic:** Heaps, Stacks and Queues

**Optimal answer from:** your app's interview approach (LeetCode #253)

**LeetCode match:** LC 253 Meeting Rooms II

Frequency: Very high | Last asked: 2026-09-29 | Roles: SWE, MLE

**Original:** https://www.1point3acres.com/interview/problems/company/google/meeting-rooms-variants

## QUESTION

LC 253 (Meeting Rooms II) appears in many forms across phone screen and onsite coding rounds — minimum number of rooms, room assignment, cyclic time, daily server jobs, gate entry/exit. Common follow-ups push toward sweep line, min-heap, or distributed processing.

The base task in every variant: given a list of intervals (start, end) — or pickup/return pairs, meeting time pairs, server job runs — answer one or more of:

Min rooms / cars / servers needed to honor all intervals concurrently (the canonical min-meeting-rooms problem).
Assignment: which interval goes to which room/car, so the caller can use the rental record.
Single-interval overlap test between two intervals.

**Variant:** parking-lot capacity

Arrival and departure intervals represent cars occupying a parking lot. Compute the peak simultaneous occupancy using the same interval-overlap model.

**Variant:** cyclic time (24h wrap)

Intervals wrap around midnight; if an interval ends after the day boundary, it carries into the next day.
**Follow-up:** "if N is huge, how do you optimize?" — expect a sweep-line discussion.

**Variant:** daily-recurring server jobs

Same set of jobs runs every day. If a job can't finish in one day, truncate at midnight and resume next day.
A server can run multiple identical jobs that overlap on the same machine; different jobs need separate servers.
**Follow-up:** a single job that runs many days needs its own dedicated server.

**Variant:** on-call rotation consolidation

## QUESTION

**Input:** on-call rotations {name, start, end} using half-open intervals.
Output maximal contiguous segments during which the active on-call set does not change; omit gaps with nobody on call.
Segments are sorted by start, and names in each segment are the active set for that time range.
Clean approach: sort starts, keep a min-heap by end time, and emit a segment whenever the next start or next end changes the active set.

**Variant:** gate entry / exit with priority rules

Array of (timestamp, action) where action in {enter, exit}; each index is a different person.
Return for each person what time they actually enter/exit, applying tie-break rules:

If the prior tick had an enter, enter wins on a tie.
If the prior tick had an exit, exit wins on a tie.
If the prior tick had no activity, exit wins.
Within the winning class, lower index wins.

Optimization: instead of stepping time tick-by-tick, jump through the sorted input timestamps to reach O(n).

**Variant:** Google Calendar free-time intersection

**Input:** requested attendees plus each person's busy calendar intervals.
Return every time range where all requested attendees are free.
Normalize each person's busy blocks into free gaps, then intersect the candidate ranges across attendees.

**Variant:** car-rental assignment

**Input:** rental orders containing an order ID, pickup time, and return time.
Assign every order to a car so that no assigned intervals overlap, minimize the number of cars, and return each car's complete order list. Model each car explicitly with its own rental record.
A current output variant instead returns a dictionary from each reservation ID to its assigned car ID; the follow-up asks for the final fleet size.
A return at time t and another pickup at time t may reuse the same car.
**Follow-up:** if the fleet is capped at K cars and all requests cannot be served, define how the allocator detects and handles the failure.

**Variant:** exact-concurrency intervals

Compute the maximum number of simultaneously active meetings.
Given a target concurrency X, also return every maximal time range during which exactly X meetings are active.
Define the event ordering when one meeting starts at the same timestamp another ends; this choice controls whether a zero-width overlap is counted.

**Variant:** maximum-profit job scheduling

## QUESTION

Each job has startTime, endTime, and profit. Choose a set of non-overlapping jobs maximizing total profit. The candidate identified it as LC 1235 (Maximum Profit in Job Scheduling). Asked as the first question of an L4 onsite round, followed by an unrelated range-query problem in the same slot.

**On-call employee metadata variant**

Each duty record includes employee ID, employee name, start time, and end time. Return the employee count and identifying information for each time segment.

EXAMPLES

```
Canonical min-meeting-rooms example:
    [[0,30],[5,10],[15,20]] -> 2.
On-call consolidation: A[10,50], B[20,60], C[30,40],
    D[30,40] -> [10,20]: A, [20,30]: A,B, [30,40]:
    A,B,C,D, [40,50]: A,B, [50,60]: B.
Single-interval overlap:
    intervalsOverlap([1,5],[4,7]) -> true.
```

Long-running job: a 36-hour daily job is impossible under the truncate rule; reserve a separate server.

## APPROACH

Drill three solutions cold: sweep-line / prefix delta, min-heap on end time, sort + greedy room assignment.
Be ready to convert between the three when interviewers add follow-ups ("give me the assignment", "now handle cyclic days").
Prepare a 1-minute pitch on map-reduce style sharding (per-shard intervals -> merged timeline -> global max).

## HOW TO THINK ABOUT IT

The number of rooms needed is the largest number of intervals overlapping at any moment. Turn each interval into +1 at its start and -1 at its end, sort the events (ends before starts at the same time) and keep a running count. To say which room each meeting gets, keep a min-heap of (end time, room) and reuse the room that frees up first.

**Basis:** LeetCode 253, Meeting Rooms II, with the common variants: room assignment and cyclic (24-hour) time.

## BRUTE FORCE

BRUTE FORCE · PSEUDO-CODE

```
def brute(intervals):
    best = 0
    for start, _ in intervals:
        busy = 0
        for s, e in intervals:
            if s <= start < e:
                busy += 1
        best = max(best, busy)
    return best
```

| TIME | SPACE |
|---|---|
| O(n^2) | O(1) |

## BRUTE FORCE

WHY IT IS SLOW

Every interval is compared with every other, O(n^2). Sorting start and end events and keeping a running count is O(n log n).

## OPTIMAL SOLUTION · Sweep line over start and end events

| TIME | SPACE |
|---|---|
| O(n log n) | O(n) |

**Code:** see FULL INTERVIEW CODE below (the optimal solution is its main function).

## FOLLOW-UP 1 Meetings wrap around midnight (cyclic time).

Split every wrapped meeting [s, e) with s >= e into [s, 24h) and [0, e) and run the same sweep: the overlap at any clock time is unchanged. A meeting that is exactly 24 hours long covers the whole day.

## FOLLOW-UP 2 The input is huge and spread over many machines.

Each shard sorts its events and produces a compressed timeline of (time, delta) changes; merge the sorted timelines with a k-way merge and take the running maximum, or add the per-shard counts at fixed time buckets. The answer is a max over time of a sum over shards.

## FULL INTERVIEW CODE · base question + every follow-up

**One program for the whole question:** the base answer and every follow-up above, built from small helper functions, with a comment above each step. Type it top to bottom in the interview; the demo at the bottom runs it.

```
import heapq
from collections import Counter


# BRUTE FORCE (say it first): Count the overlap at every
# start time | time O(n^2)
def min_rooms_brute(intervals):
    return max(
        (
            sum(
                1
                for s2, e2 in intervals
                if s2 <= s < e2
            )
            for s, e in intervals
        ),
        default=0,
    )


# BASE QUESTION (optimal): Sweep line over start and end
# events
def min_rooms(intervals):
    # +1 at every start, -1 at every end
    events = []
    for s, e in intervals:
```

```
        events.append((s, 1))
        events.append((e, -1))
    # at equal times ends (-1) sort before starts (+1)
    events.sort()
    cur = best = 0
    for _, delta in events:
        cur += delta
        best = max(best, cur)
    return best


# VARIANT: which room for each meeting
def assign_rooms(intervals):
    # which room does each meeting get?
    order = sorted(range(len(intervals)),
                   key=lambda i: intervals[i])
    heap, rooms, used = [], [0] * len(intervals), 0
    for i in order:
        start, end = intervals[i]
```

```
        # reuse the room that frees first, if it is free
        if heap and heap[0][0] <= start:
            _, room = heapq.heappop(heap)
        else:
            room, used = used, used + 1
        rooms[i] = room
        heapq.heappush(heap, (end, room))
    return rooms


# VARIANT: on-call consolidation
def oncall_segments(rotations):
    # rotations: (name, start, end), half-open intervals
    delta = {}
    for name, s, e in rotations:
        delta.setdefault(s, []).append((name, 1))
        delta.setdefault(e, []).append((name, -1))
    active, out, times = Counter(), [], sorted(delta)
    for t, nxt in zip(times, times[1:]):
        # apply every change that happens at time t
        for name, d in delta[t]:
            active[name] += d
        who = frozenset(n for n in active if active[n])
        # skip gaps; merge with the previous equal segment
        if not who:
            continue
        if out and out[-1][1] == t and out[-1][2] == who:
            out[-1] = (out[-1][0], nxt, who)
        else:
            out.append((t, nxt, who))
    return out


# FOLLOW-UP 1: Meetings wrap around midnight (cyclic
# time).
def min_rooms_cyclic(intervals, day=24 * 60):
    parts = []
    for s, e in intervals:
        if s < e:
            parts.append((s, e))
        else:
            # wraps past midnight: split at the boundary
            parts += [(s, day), (0, e)]
    return min_rooms(parts)
```

```
if __name__ == "__main__":
    base = [(0, 30), (5, 10), (15, 20)]
    assert min_rooms(base) == 2
    # touching meetings share a room
    assert min_rooms([(5, 8), (8, 9)]) == 1
    rooms = assign_rooms(base)
    assert rooms[0] != rooms[1] and rooms[1] == rooms[2]
    wrap = [(22 * 60, 2 * 60), (23 * 60, 60), (0, 3 * 60)]
    assert min_rooms_cyclic(wrap) == 3
    rot = [("A", 1, 5), ("B", 3, 8)]
    assert oncall_segments(rot) == [
        (1, 3, frozenset("A")),
        (3, 5, frozenset("AB")),
        (5, 8, frozenset("B"))]
    # a gap with nobody on call is left out
    gap = [("A", 1, 2), ("B", 4, 5)]
    assert len(oncall_segments(gap)) == 2
```

## NOTES FROM CANDIDATES

Asked in Round 1 coding, Round 2 onsite coding, phone screen, and as a warm-up in the FDE loop. Almost every other coding loop contains some flavor of this family.

Interviewer commonly chains two follow-ups: (1) reconstruct assignment, (2) scale out to many shards / regions.

For the assignment follow-up, min-heap keyed on end time is the clean answer: pop the earliest-ending room, push the new interval.

For "distributed across shards", expect open-ended discussion: per-shard sweep, cross-shard merge, reconciliation; the interviewer often acknowledges the question is intentionally vague.

The on-call metadata variant includes detailed complexity follow-ups: account for heap operations, stored employee information, and output size separately, and explain each symbol in the complexity expression.

## PREPARATION

Drill three solutions cold: sweep-line / prefix delta, min-heap on end time, sort + greedy room assignment.

Be ready to convert between the three when interviewers add follow-ups ("give me the assignment", "now handle cyclic days").

Prepare a 1-minute pitch on map-reduce style sharding (per-shard intervals -> merged timeline -> global max).

**STUDY HERE: topic 4, Heaps, Stacks and Queues**

Also fits: Binary Search and Sorting, Arrays, Hashing and Two Pointers, Design, Streaming and Scale (listed there as SKIP, no pages). Site tags: hashmap, set, sorting, top-k, streaming, medium

**MEDIUM | 1Point3Acres | phone-screen | 4 of 11**

# Active Communication Top Users

**ASKED 2026-09-22 | 28 DAYS AGO | LAST 30 DAYS**

**STUDY: READ** **NO LEETCODE FREQUENCY** **PRIORITY 228/228**

## ABOUT THIS QUESTION

**Topic:** Heaps, Stacks and Queues

**Optimal answer from:** written and tested here for this exact question

Frequency: Low | Last asked: 2026-09-22 | Roles: SWE

**Original:** https://www.1point3acres.com/interview/problems/company/google/active-communication-top-users

## QUESTION

Process a continuous stream of user pairs such as **`A-B`** and **`B-C`**; each pair creates an active communication edge, and the system should output users ordered by current communication count. Follow-ups add self-communication handling, duplicate communication dedupe, and top-K users by count. An additional API variant exposes register and mostActiveUser(), then asks for constant-time most-active lookup and a changing-K ranking extension.SWEhashmapsetsortingtop-kstreamingmediumFrequencyLowLast asked2026-09-22Stagephone-screen

## APPROACH

Implement the pair-normalization layer separately from count maintenance.
Practice maintaining user -> count, edge set, and count -> ordered users views without letting updates drift.
Dry-run duplicate edges and self-edges before discussing top-K optimization.

## INTERVIEW-DAY PLAN

### CLARIFY FIRST

Ask whether a pair A-B counts for both users, whether repeated pairs count once, how self communication is treated, how ties order, and whether k changes over time.

### BRUTE FORCE TO SAY OUT LOUD

Recount from all edges and sort users on every query. O(E log U).

### OPTIMAL IDEA

Normalize pairs (sorted tuple). Keep an edge set for deduplication, a dict of user to count, and either a sorted structure of (count, user) or a bucket map count to ordered users so an update moves a user by one bucket. Top-K reads the highest buckets. Note the edge set can use quadratic space in the number of users; say so.

### DRY-RUN THESE CASES

## INTERVIEW-DAY PLAN

A-B then B-A (duplicate), a self pair, ties on count, a new user, k larger than the number of users.

### LIKELY FOLLOW-UPS

Constant-time most active user (track the maximum count with buckets), a changing k, or expiry of old edges.

### PITFALLS

Claiming linear space while storing all pairs. Counting a duplicate edge twice.

## HOW TO THINK ABOUT IT

Normalize the pair first (sorted tuple) so A-B and B-A are one edge, then keep an edge set for dedup and a count per user. The ranking part is a separate question: top-K comes from a heap, and the single most active user can be maintained in O(1) because counts only ever go up. Mention that the edge set uses quadratic space in the worst case.

**Basis:** Written. Assumed rules (confirm them): a pair 'A-B' is an unordered communication edge; repeated pairs count once; a user's activity is the number of distinct partners; order by activity descending, then name; self pairs (A-A) are ignored unless a flag says otherwise.

## BRUTE FORCE

### BRUTE FORCE · PSEUDO-CODE

```
pairs = set()
def add(a, b):
    pairs.add((min(a, b), max(a, b)))
def top_users():
    partners = defaultdict(set)
    for a, b in pairs:
        partners[a].add(b)
        partners[b].add(a)
    def rank(u):
        return (-len(partners[u]), u)
    return sorted(partners, key=rank)
```

| TIME | SPACE |
|---|---|
| **O(E + U log U)** per query | **O(E)** |

### WHY IT IS SLOW

Each query rebuilds all counts and sorts, O(E + U log U). Maintaining counts incrementally with count buckets gives O(1) updates and an O(1) most-active lookup.

## OPTIMAL SOLUTION · Edge set, counts per user, count buckets for the maximum

| TIME | SPACE |
|---|---|
| **O(1)** per pair, O(U log k) for top-K, O(1) for the most active user | **O(E)** |

**Code:** see FULL INTERVIEW CODE below (the optimal solution is its main function).

**FOLLOW-UP 1 Self communication (A-A) should now count.**

Pass a flag to the class: a self pair is one edge that adds one to that user's count (and only once). The dedup set already handles repeated self pairs.

**FOLLOW-UP 2 Top-K with a changing K, or a fixed-size priority queue.**

A heap of size K can be wrong after counts change, because an element outside it can overtake one inside. Re-run heapq.nsmallest on the count dict per query (O(U log K)) or maintain a sorted structure keyed by (count, name) updated on every increment.

**FOLLOW-UP 3 The space claim.**

Distinct pairs can number U^2 / 2 in the worst case, so the edge set is quadratic in the users; the counts alone are linear. If duplicates are not a concern, drop the set and keep O(U).

**FULL INTERVIEW CODE · base question + every follow-up**

**One program for the whole question:** the base answer and every follow-up above, built from small helper functions, with a comment above each step. Type it top to bottom in the interview; the demo at the bottom runs it.

```python
import heapq
from collections import Counter, defaultdict


# BRUTE FORCE (say it first): Recompute counts from all
# edges on every query | time O(E + U log U) per query
def _norm(pair):
    a, b = pair.split("-")
    return (a, b) if a <= b else (b, a)


class ActiveBrute:
    def __init__(self, count_self=False):
        self.edges, self.count_self = set(), count_self

    def add(self, pair):
        a, b = _norm(pair)
        if a == b and not self.count_self:
            return
        self.edges.add((a, b))

    def counts(self):
        c = Counter()
        for a, b in self.edges:
            c[a] += 1
            if a != b:
                c[b] += 1
        return c

    def top_k(self, k):
        c = self.counts()
        return sorted(c, key=lambda u: (-c[u], u))[:k]

    def most_active(self):
        c = self.counts()
        if not c:
            return set()
        m = max(c.values())
        return {u for u, v in c.items() if v == m}
```

```python
# BASE QUESTION (optimal): Edge set, counts per user,
# count buckets for the maximum
# FOLLOW-UP 1: Self communication (A-A) should now count.
class Active:
```

```python
    def __init__(self, count_self=False):
        # distinct edges seen so far (for the dedupe rule)
        self.edges = set()
        # user -> how many different partners
        self.count = Counter()
        # count -> users having exactly that count
        self.bucket = defaultdict(set)
        # the highest count right now (it never drops)
        self.top = 0
        self.count_self = count_self

    def bump(self, user):
        c = self.count[user]
        # move the user from bucket c to bucket c + 1
        if c:
            self.bucket[c].discard(user)
        self.count[user] = c + 1
        self.bucket[c + 1].add(user)
        self.top = max(self.top, c + 1)

    def add(self, a, b):
        # A-B and B-A are the same edge
        pair = (min(a, b), max(a, b))
        # follow-ups: skip self pairs (unless counted)
        # and skip a repeated pair
        skip_self = a == b and not self.count_self
        if skip_self or pair in self.edges:
            return
        self.edges.add(pair)
        self.bump(a)
        # a self pair counts once, not twice
        if a != b:
            self.bump(b)

    def most_active(self):
        # O(1): read the top bucket
        if not self.top:
            return set()
        return set(self.bucket[self.top])

    def top_k(self, k):
        # k users by count, ties broken by name
        def rank(u):
            return (-self.count[u], u)
        return heapq.nsmallest(k, self.count, key=rank)
```

```python
if __name__ == "__main__":
    a = Active()
    for x, y in [("A", "B"), ("B", "C"), ("B", "A")]:
        a.add(x, y)
    # B talks to A and C; the repeat B-A is ignored
    assert a.most_active() == {"B"}
    assert a.count["B"] == 2 and a.count["A"] == 1
    assert a.top_k(2) == ["B", "A"]
    # self pairs are ignored by default
    a.add("C", "C")
    assert a.count["C"] == 1
    # ... but can be switched on, counting once
    s = Active(count_self=True)
    s.add("C", "C")
    s.add("C", "C")
    assert s.count["C"] == 1
    assert Active().most_active() == set()
```

## NOTES FROM CANDIDATES

A Java solution used an ordered map / tree structure to keep counts sortable.
The interviewer asked many clarification questions and expected dry runs.
The first version did not require self-communication handling; it was added as a coded follow-up.
Account for the per-user contact sets when explaining space use: the distinct relationship state can take quadratic space in the number of users. The interviewer explicitly challenged an initial linear-space claim.
A fixed-size priority-queue proposal did not finish the changing-K discussion; the interviewer asked for further optimization.
Maintaining the API variant
With registrations only and any tied maximum acceptable, normalize undirected pairs and store the distinct edges. For a new edge, increment each distinct endpoint once; an accepted self-edge changes one count, while an ignored self-edge changes none. Clarify that choice and the empty-state response.
Cache a user with the largest count. Counts never decrease, so only the endpoints of a new edge can displace the cached winner; duplicate edges do not change it. Reading this cache makes mostActiveUser() O(1), with expected O(1) registration work using

hash-based edge membership and counters under unit-cost user identifiers. A stricter tie rule requires separate handling.
For changing K, one option is a balanced ordered index of all users keyed by count and an agreed tie-breaker. Remove and reinsert the affected users when counts change. Updates cost O(log U); starting at the highest-ranked user and walking entries in descending-count order costs O(log U + K) for at most U returned users. This supports different K values without rebuilding a fixed-capacity heap and is a preparation option, not a specified interviewer solution.
Counters, the maximum cache, and the ranking index use O(U) space. Distinct relationship storage adds O(E), giving O(U + E) overall and O(U^2) in a dense graph.

## PREPARATION

Implement the pair-normalization layer separately from count maintenance.
Practice maintaining user -> count, edge set, and count -> ordered users views without letting updates drift.
Dry-run duplicate edges and self-edges before discussing top-K optimization.

**STUDY HERE: topic 4, Heaps, Stacks and Queues**

Also fits: Design, Streaming and Scale (listed there as SKIP, no pages). Site tags: iterator, queue, data-structure, object-design, medium

MEDIUM | 1Point3Acres | onsite-coding | 5 of 11

# Implement Python's itertools.tee

**ASKED 2026-08-20 | 61 DAYS AGO | LAST 90 DAYS**

**STUDY: READ** | **NO LEETCODE FREQUENCY** | **PRIORITY 225/228**

## ABOUT THIS QUESTION

**Topic:** Heaps, Stacks and Queues

**Optimal answer from:** written and tested here for this exact question

Frequency: Single report | Last asked: 2026-08-20 | Roles: SWE

**Original:** https://www.1point3acres.com/interview/problems/company/google/python-tee-independent-iterators

## QUESTION

Implement the behavior of Python's `tee`: accept one iterator and an integer **n**, then return **n** independent iterators over the same source sequence. Define the required function and classes yourself and improve a naive design toward an efficient shared-state implementation.SWEiteratorqueuedata-structureobject-designmediumFrequencySingle reportLast asked2026-08-20Stageonsite-coding

## APPROACH

Implement a custom Python iterator class from memory, including __iter__ and __next__.
Trace two and three consumers advancing at different speeds and identify what state must remain available.
Compare a fully materialized baseline with a lazy shared-state design and explain the memory behavior.

## INTERVIEW-DAY PLAN

### CLARIFY FIRST

Ask whether the source iterator can be consumed by one consumer first, what happens if one consumer is far ahead, and whether thread safety is needed.

### BRUTE FORCE TO SAY OUT LOUD

Store the whole source in a list and give each consumer an index. Simple, but defeats the purpose for large or infinite iterators.

### OPTIMAL IDEA

Keep a deque per consumer. When a consumer needs an item and its deque is empty, pull the next item from the source and append it to every consumer's deque. Each item is held only until all consumers have read it. Build with a shared state object and a small iterator class with __iter__ and __next__. The linked-list variant uses one chain of nodes with each consumer holding a pointer, which frees nodes as the slowest consumer moves.

### DRY-RUN THESE CASES

Two consumers at different speeds, n = 1, n = 0, an empty source, an exhausted source.

### LIKELY FOLLOW-UPS

Thread safety (a lock around the pull), and memory growth if one consumer stalls.

### PITFALLS

Pulling from the source once per consumer instead of once per item. Not appending to the other deques.

## HOW TO THINK ABOUT IT

The source can only be consumed once, so someone must remember items until every consumer has read them. The state to keep is exactly 'items the slowest consumer has not read yet', so either give each consumer a queue that the leader fills, or keep one shared linked list with a pointer per consumer so memory is freed automatically.

**Basis:** The itertools.tee documentation's recipe; the linked-list variant is how CPython shares buffers.

## BRUTE FORCE

### BRUTE FORCE · PSEUDO-CODE

```
def tee(source, n):
    items = list(source)  # read it all now
    def reader():
        for x in items:
            yield x
    return [reader() for _ in range(n)]
```

| TIME | SPACE |
|---|---|
| **O(n)** up front | **O(n)** |

### WHY IT IS SLOW

It reads the source completely before anything is consumed, never works on an infinite or lazy source, and keeps every item in memory. One buffer per consumer (or a shared linked list) holds only what the slowest consumer has not read.

## OPTIMAL SOLUTION · One deque per consumer, filled lazily

| TIME | SPACE |
|---|---|
| **O(n)** total, O(1) amortized per item per consumer | **O(lag between fastest and slowest consumer)** |

**Code:** see FULL INTERVIEW CODE below (the optimal solution is its main function).

### FOLLOW-UP 1 Memory grows when one consumer stalls. Is there a way to free items automatically?

Use one shared singly linked list with a pointer per consumer. The only references to old nodes are the consumers' pointers, so Python's garbage collector frees nodes behind the slowest consumer without any bookkeeping. This is the design CPython uses.

### FOLLOW-UP 2 Make it safe for several threads.

Pull from the source under one lock and append to the queues inside it. Without the lock two consumers can pull two items and append them in different orders to different queues. **Cost:** one lock acquisition per item pulled, O(1) extra per item; contention grows with the number of consumers.

### FOLLOW-UP 3 Why not just copy the iterator?

Most iterators (files, generators, sockets) cannot be copied. tee gives independent views by buffering, which is the only thing you can do when the source cannot rewind.

### FULL INTERVIEW CODE · base question + every follow-up

**One program for the whole question:** the base answer and every follow-up above, built from small helper functions, with a comment above each step. Type it top to bottom in the interview; the demo at the bottom runs it.

```python
import threading
from collections import deque


# FIRST IDEA (simplest): copy into a list
def tee_naive(iterable, n=2):
    # simplest idea: copy everything into a list
    items = list(iterable)
    # but it never works for endless iterators
    return tuple(iter(items) for _ in range(n))


# BASE QUESTION (optimal): One deque per consumer, filled
# lazily
def tee_queues(iterable, n=2):
    source = iter(iterable)
    # one waiting line of values per consumer
    queues = [deque() for _ in range(n)]

    def consumer(mine):
        while True:
            if not mine:
                try:
                    value = next(source)
                except StopIteration:
                    return
                # give the new value to everybody
                for q in queues:
                    q.append(value)
            yield mine.popleft()

    return tuple(consumer(q) for q in queues)


# FOLLOW-UP 1: Memory grows when one consumer stalls. Is
# there a way to free items ...
class Cell:
    # one link of the shared list
    def __init__(self):
        self.filled = False
        self.value = None
        self.next = None


# FOLLOW-UP 2: Make it safe for several threads.
def tee(iterable, n=2):
    source = iter(iterable)
    # one lock keeps the pulls in a single order
    lock = threading.Lock()

    def consumer(cell):
        while True:
            with lock:
                if not cell.filled:
                    # the first reader pulls the value
                    try:
                        cell.value = next(source)
                    except StopIteration:
                        return
                    cell.filled = True
                    cell.next = Cell()
                value, cell = cell.value, cell.next
            yield value

    # every reader starts at the same first cell
    head = Cell()
    return tuple(consumer(head) for _ in range(n))


if __name__ == "__main__":
    a, b = tee(range(5))
    # readers are independent, at any speed
    assert next(a) == 0 and next(a) == 1
    assert list(b) == [0, 1, 2, 3, 4]
    assert list(a) == [2, 3, 4]
    x, y, z = tee_queues("abc", 3)
    assert list(x) == list(y) == list(z) == list("abc")
    # works on an endless source, which tee_naive cannot
    def count():
        i = 0
        while True:
            yield i
            i += 1
    e1, e2 = tee(count())
    assert [next(e1) for _ in range(3)] == [0, 1, 2]
    assert next(e2) == 0
    assert [list(i) for i in tee_naive([1, 2], 2)] == [
        [1, 2], [1, 2]]
```

## NOTES FROM CANDIDATES

The round depends on Python iterator-protocol fluency as much as general data-structure reasoning. Clarify laziness, source exhaustion, invalid n, and whether concurrent consumption is in scope before implementing.
A correct baseline consumes the source lazily once and gives each returned iterator its own queue: when one queue is empty, pull one source value, append it to every queue, then return it from the requesting queue. If n iterators trail across a gap of g values, however, those queues can retain O(n·g) buffered copies. For the optimal shared-state design, keep one linked buffer (or indexed deque) of source values plus one cursor per iterator, and discard a buffered value only after every cursor has passed it. That preserves independent consumption with O(g+n) auxiliary state, where g is the distance between the fastest and slowest consumers.

## PREPARATION

Implement a custom Python iterator class from memory, including __iter__ and __next__.
Trace two and three consumers advancing at different speeds and identify what state must remain available.
Compare a fully materialized baseline with a lazy shared-state design and explain the memory behavior.

**STUDY HERE: topic 4, Heaps, Stacks and Queues**

Also fits: Arrays, Hashing and Two Pointers, Design, Streaming and Scale (listed there as SKIP, no pages). Site tags: interval, two-pointer, heap, streaming, medium

MEDIUM | 1Point3Acres | onsite-coding | 6 of 11

# Union of Sorted Interval Lists

ASKED 2026-08-16 | 65 DAYS AGO | LAST 90 DAYS

STUDY: TALK | ALL TIME 25.5 | 30 DAYS 0 | PRIORITY 139/228

## ABOUT THIS QUESTION

**Topic:** Heaps, Stacks and Queues

**Optimal answer from:** written and tested here for this exact question

**Related in your app (a neighbouring problem, not this exact question):** LeetCode #986 Interval List Intersections

**LeetCode match:** LC 986 Interval List Intersections (related)

Frequency: Single report | Last asked: 2026-08-16 | Roles: SWE

**Original:** https://www.1point3acres.com/interview/problems/company/google/sorted-interval-list-union

## QUESTION

Reverse of LC 986: given two individually sorted, internally non-overlapping interval lists, return their union. Follow-ups extend to K lists with a heap, inputs too large for memory, and lists that may overlap internally.SWEintervaltwo-pointerheapstreamingmediumFrequencySingle reportLast asked2026-08-16Stageonsite-coding

## APPROACH

Code the two-list merge without special-casing more than an empty result and a non-overlapping next interval.
Generalize the implementation to K iterators using a heap, then explain the memory bound.
Rehearse how the invariant changes when individual inputs are not internally disjoint.

## INTERVIEW-DAY PLAN

### CLARIFY FIRST

Ask whether intervals are closed or half-open, whether touching intervals merge, whether each list is sorted and non-overlapping internally (later follow-up removes this), and how large the inputs are.

### BRUTE FORCE TO SAY OUT LOUD

Concatenate both lists, sort, and merge. O((m + n) log (m + n)).

### OPTIMAL IDEA

Two-pointer merge: repeatedly take the interval with the smaller start from either list and fold it into the last output interval if it overlaps (start at most the previous end), else append it. O(m + n). For K lists use heapq.merge (a heap of K heads), and since it is a generator the inputs never need to fit in memory.

### DRY-RUN THESE CASES

One list empty, no overlaps, one interval swallowing many, touching endpoints, identical lists.

## INTERVIEW-DAY PLAN

### LIKELY FOLLOW-UPS

K lists, lists too big for memory, lists that overlap internally (sort each first or heap everything).

### PITFALLS

Using end instead of start to pick the next interval. Not taking the maximum of ends when merging.

## HOW TO THINK ABOUT IT

Each list is already sorted, so you can walk both with two pointers like the merge step of merge sort: always take the interval with the smaller start next. Once the intervals come out in start order, merging is the usual one-pass fold: extend the last output interval if the next one overlaps or touches it, otherwise start a new one.

**Basis:** The reverse of LeetCode 986 (Interval List Intersections): union instead of intersection.

## BRUTE FORCE

### BRUTE FORCE · PSEUDO-CODE

```
def brute(list1, list2):
    merged = []
    for s, e in sorted(list1 + list2):
        if merged and s <= merged[-1][1]:
            merged[-1][1] = max(merged[-1][1], e)
        else:
            merged.append([s, e])
    return merged
```

| TIME | SPACE |
|---|---|
| O((m + n) log (m + n)) | O(m + n) |

### WHY IT IS SLOW

Sorting ignores that both lists are already sorted: O((m + n) log (m + n)) instead of O(m + n) for a two-pointer merge.

## OPTIMAL SOLUTION · Two-pointer merge

| TIME | SPACE |
|---|---|
| O(m + n) | O(m + n) for the output |

**Code:** see FULL INTERVIEW CODE below (the optimal solution is its main function).

## FOLLOW-UP 1 What if the lists are too big for memory?

Read each list through an iterator (a file or a paged API). union_k only ever holds one interval per list plus the current merged interval, so memory is O(K) and it can emit results as it goes. Sorted order is what makes streaming possible.

**FOLLOW-UP 2 What if one list may overlap itself?**

The two-pointer merge already works, because it folds every incoming interval into the last output interval. What breaks is only the assumption that each list is sorted by start; if that also fails, sort each list first (O(n log n)) or push everything through a heap.

**FOLLOW-UP 3 Do touching intervals merge?**

Decide the endpoint rule before coding. For closed intervals [1, 3] and [3, 5] share a point, so they merge (start <= last end). For half-open intervals [1, 3) and [3, 5) they do not overlap but are adjacent; whether to merge adjacent ones is a product choice, so ask.

**FOLLOW-UP 4 Each list may overlap itself or be unsorted.**

Normalise each list first (sort and merge it by itself, O(n log n)), after which the two-pointer union applies. Or sort everything together if memory allows.

**FULL INTERVIEW CODE · base question + every follow-up**

**One program for the whole question:** the base answer and every follow-up above, built from small helper functions, with a comment above each step. Type it top to bottom in the interview; the demo at the bottom runs it.

```
import heapq


# BRUTE FORCE (say it first): Concatenate, sort and merge
# | time O((m + n) log (m + n))
def union_sort(a, b):
    out = []
    for s, e in sorted(a + b):
        if out and s <= out[-1][1]:
            out[-1][1] = max(out[-1][1], e)
        else:
            out.append([s, e])
    return out


# HELPERS for the base question
# NEEDED BY THE BASE QUESTION: it is also the answer to
# the follow-up below
# FOLLOW-UP 1: What if the lists are too big for memory?
def union_stream(lists, touching=True):
    # lists: sorted iterables of (start, end)
    merged = heapq.merge(*lists)
    cur = None
    for start, end in merged:
        if cur is None:
            cur = [start, end]
            continue
        # touching: [1, 3] and [3, 5] share the point 3
        joined = start < cur[1]
        if touching:
            joined = start <= cur[1]
        if joined:
            cur[1] = max(cur[1], end)
        else:
            # finished one block: emit it as we go
            yield tuple(cur)
            cur = [start, end]
    if cur is not None:
        yield tuple(cur)


# BASE QUESTION (optimal): Two-pointer merge
def union_two(a, b, touching=True):
    return list(union_stream([a, b], touching))


# VARIANT: K lists with a heap
def union_k(lists, touching=True):
    # same code, any number of lists, O(K) memory
    return list(union_stream(lists, touching))


# FOLLOW-UP 2: What if one list may overlap itself?
def normalise(intervals):
    # a list that is unsorted or overlaps itself
    ordered = sorted(intervals)
    return list(union_stream([ordered]))


# FOLLOW-UP 4: Each list may overlap itself or be
# unsorted.
def union_unsorted(a, b):
    # fix each list first, then use the same union
    return union_two(normalise(a), normalise(b))


if __name__ == "__main__":
    a = [(1, 3), (6, 8)]
    b = [(2, 4), (9, 10)]
    assert union_two(a, b) == [(1, 4), (6, 8), (9, 10)]
    # closed intervals that touch merge
    assert union_two([(1, 3)], [(3, 5)]) == [(1, 5)]
    # half-open intervals that only touch stay apart
    assert union_two([(1, 3)], [(3, 5)], False) == [
        (1, 3), (3, 5)]
    # three lists, one of them empty
    three = union_k([a, b, []])
    assert three == [(1, 4), (6, 8), (9, 10)]
    # a list that overlaps itself, and an unsorted one
    merged = union_unsorted([(5, 9), (1, 6)], [(8, 12)])
    assert merged == [(1, 12)]
    assert union_unsorted([(9, 10), (1, 2)], []) == [
        (1, 2), (9, 10)]
```

**NOTES FROM CANDIDATES**

The round intentionally layers one constraint at a time. Keep the base invariant explicit—every interval before the current pointer has already been absorbed into a canonical output prefix—so the streaming and K-way versions remain easy to justify.
Agree on endpoint semantics before coding: touching closed intervals merge, while touching half-open intervals do not overlap. For K streaming inputs, the heap plus one current interval per input uses O(K) working memory, excluding the output.

**PREPARATION**

Code the two-list merge without special-casing more than an empty result and a non-overlapping next interval.
Generalize the implementation to K iterators using a heap, then explain the memory bound.
Rehearse how the invariant changes when individual inputs are not internally disjoint.

**STUDY HERE: topic 4, Heaps, Stacks and Queues**

Also fits: Design, Streaming and Scale (listed there as SKIP, no pages). Site tags: heap, streaming, medium

MEDIUM | IN APP LC 295 | 1Point3Acres | tech-screen | 7 of 11

# Streaming Median (LC 295) with Follow-ups

ASKED 2026-08-03 | 78 DAYS AGO | LAST 90 DAYS

STUDY: TYPE | ALL TIME 50.4 | 30 DAYS 0 | PRIORITY 69/228

## ABOUT THIS QUESTION

**Topic:** Heaps, Stacks and Queues

**Optimal answer from:** your app's interview approach (LeetCode #295)

**LeetCode match:** LC 295 Find Median from Data Stream

Frequency: Single report | Last asked: 2026-08-03 | Roles: SWE

**Original:** https://www.1point3acres.com/interview/problems/company/google/streaming-median-with-follow-ups

## QUESTION

A near-canonical LC 295 streaming-median implementation using two heaps, followed by two design extensions: exploit a very small value domain, then support an arbitrary k-th-smallest query instead of only the median.SWEheapstreamingmediumFrequencySingle reportLast asked2026-08-03Stagetech-screen

## APPROACH

Implement the two-heap median structure and dry-run odd and even stream lengths.
Sketch a bounded-domain alternative and state its update and query costs.
Compare what must change when the requested order statistic is an arbitrary k rather than a fixed median.

## INTERVIEW-DAY PLAN

### CLARIFY FIRST

Ask whether values are integers, the range of values, how often median queries happen compared with inserts, and what to return for an even count (average of two middles).

### BRUTE FORCE TO SAY OUT LOUD

Keep a sorted list and insert with bisect; median is the middle. O(n) per insert due to shifting.

### OPTIMAL IDEA

Two heaps: a max-heap for the lower half and a min-heap for the upper half, sizes within one. Insert by pushing through one heap into the other and rebalancing. Median is the top of the larger heap or the average of both tops. O(log n) insert, O(1) median.

### DRY-RUN THESE CASES

One element, two elements, duplicates, all equal, alternating very large and very small values.

## INTERVIEW-DAY PLAN

### LIKELY FOLLOW-UPS

Values in a small range (counting array), arbitrary k-th smallest (Fenwick tree or order-statistic tree), or too much data (quantile sketch).

### PITFALLS

Rebalancing after the wrong heap push. Integer division when the median should be fractional.

## HOW TO THINK ABOUT IT

Keep the data split at the median: a max-heap for the smaller half and a min-heap for the larger half, with sizes equal or differing by one. Then the median is read from the tops. Insert by pushing through one heap into the other to keep every smaller value on the left, and rebalance the sizes.

**Basis:** LeetCode 295, Find Median from Data Stream.

## BRUTE FORCE

### BRUTE FORCE · PSEUDO-CODE

```
nums = []  # always kept sorted
def add(x):
    insort(nums, x)
def median():
    n = len(nums)
    if n % 2:
        return nums[n // 2]
    return (nums[n // 2 - 1] + nums[n // 2]) / 2
```

| TIME | SPACE |
|---|---|
| O(n) per insert (list shift), O(1) median | O(n) |

### WHY IT IS SLOW

Each insert shifts up to n elements, O(n) per add (fast in practice, but linear). Two heaps keep the median available with O(log n) per add and O(1) per query.

## OPTIMAL SOLUTION · Two heaps

| TIME | SPACE |
|---|---|
| O(log n) add, O(1) median | O(n) |

**Code:** see FULL INTERVIEW CODE below (the optimal solution is its main function).

## FOLLOW-UP 1 Almost every value is in a small range (say 0 to 100). Can you do better than two heaps?

Yes: a counting array gives O(1) inserts, and the median needs one pass over at most V counts, so O(V) per query with V fixed. Keep a moving pointer to the median bucket to make queries amortized O(1).

**FOLLOW-UP 2 Now support an arbitrary k-th smallest, not only the median.**

Two heaps only work for one fixed rank. Use an order-statistics structure: a Fenwick tree over (compressed) values, or a balanced BST augmented with subtree sizes. Both give O(log n) add and O(log n) k-th, and the Fenwick tree is far shorter to write in an interview.

**FOLLOW-UP 3 What if the data stream is too large to store?**

Exact medians need memory proportional to the data, so use a quantile sketch (t-digest, GK, KLL). They answer any quantile within a chosen error bound in O(1/epsilon) memory.

**FOLLOW-UP 4 The median of only the last K values.**

Keep the window in a deque (to know what expires) and a sorted list (to read the middle); insert and remove with bisect. O(K) per update because of list shifting; two heaps with lazy deletion give O(log K).

**FULL INTERVIEW CODE · base question + every follow-up**

**One program for the whole question:** the base answer and every follow-up above, built from small helper functions, with a comment above each step. Type it top to bottom in the interview; the demo at the bottom runs it.

```python
import bisect
import heapq
from bisect import bisect_left, insort


# BRUTE FORCE (say it first): Keep a sorted list and
# insert with bisect | time O(n) per insert (list shift),
# O(1) median
class MedianSorted:
    def __init__(self):
        self.a = []

    def addNum(self, x):
        bisect.insort(self.a, x)

    def findMedian(self):
        n = len(self.a)
        return (
            float(self.a[n // 2])
            if n % 2
            else (self.a[n // 2 - 1] + self.a[n // 2])
            / 2
        )


# BASE QUESTION (optimal): Two heaps
class MedianFinder:
    def __init__(self):
        # low: max-heap (stored negated); high: min-heap
        self.low, self.high = [], []

    def add(self, x):
        # push through low so the halves stay ordered
        heapq.heappush(self.low, -x)
        top = -heapq.heappop(self.low)
        heapq.heappush(self.high, top)
        # keep low the same size as high, or one bigger
        if len(self.high) > len(self.low):
            top = heapq.heappop(self.high)
            heapq.heappush(self.low, -top)

    def median(self):
        if len(self.low) > len(self.high):
            return float(-self.low[0])
        return (-self.low[0] + self.high[0]) / 2


# FOLLOW-UP 1: Almost every value is in a small range (say
# 0 to 100). Can you do ...
class CountingMedian:
    """Follow-up: values are whole numbers in 0..maxv."""

    def __init__(self, maxv):
        self.counts = [0] * (maxv + 1)
        self.n = 0

    def add(self, x):
        self.counts[x] += 1
        self.n += 1

    def kth(self, k):
        # k-th smallest (1-based): one pass over the
        # counts
        seen = 0
        for value, c in enumerate(self.counts):
            seen += c
            if seen >= k:
                return value

    def median(self):
        lo = self.kth((self.n + 1) // 2)
        hi = self.kth(self.n // 2 + 1)
        return (lo + hi) / 2


# FOLLOW-UP 2: Now support an arbitrary k-th smallest, not
# only the median.
class Fenwick:
    """Follow-up: any k-th smallest in O(log V)."""

    def __init__(self, size):
        self.size, self.tree = size, [0] * (size + 1)

    def add(self, value):
        i = value + 1
        while i <= self.size:
            self.tree[i] += 1
            i += i & -i

    def kth(self, k):
        # walk down the tree to the k-th smallest value
        pos, step = 0, 1 << self.size.bit_length()
        while step:
            nxt = pos + step
            if nxt <= self.size and self.tree[nxt] < k:
                pos = nxt
                k -= self.tree[nxt]
            step >>= 1
        return pos


# FOLLOW-UP 4: The median of only the last K values.
class WindowMedian:
    """Follow-up: median of only the last k values."""

    def __init__(self, k):
        self.k, self.order, self.sorted = k, [], []

    def add(self, x):
```

```
        self.order.append(x)
        insort(self.sorted, x)
        if len(self.order) > self.k:
            old = self.order.pop(0)
            del self.sorted[bisect_left(self.sorted, old)]

    def median(self):
        s, n = self.sorted, len(self.sorted)
        return (s[(n - 1) // 2] + s[n // 2]) / 2


if __name__ == "__main__":
    m = MedianFinder()
    for x in [5, 2, 8]:
        m.add(x)
    assert m.median() == 5
    m.add(1)
    assert m.median() == 3.5
    c = CountingMedian(100)
    for x in [5, 2, 8, 1]:
        c.add(x)
    assert c.median() == 3.5
    f = Fenwick(100)
    for x in [5, 2, 8, 1]:
        f.add(x)
    ranks = [f.kth(k) for k in (1, 2, 3, 4)]
```

```
    assert ranks == [1, 2, 5, 8]
    w = WindowMedian(3)
    for x in [9, 1, 5, 7]:
        w.add(x)
    # window is [1, 5, 7]
    assert w.median() == 5
```

## NOTES FROM CANDIDATES

Keep the lower half in a max-heap and the upper half in a min-heap. Maintain a size difference of at most one and keep every lower-half value no greater than every upper-half value.
For an even-sized stream, average the two heap roots.
Each insertion costs O(log n), a median query costs O(1), and the heaps use O(n) space.
For a small bounded domain, a frequency array gives O(1) updates and an O(U) order-statistic scan, where U is the domain size. For arbitrary k-th-smallest queries, use an order-statistic tree or a Fenwick/segment tree over a known or compressed domain for logarithmic update and selection.
The base problem required complete working code plus a dry run and complexity analysis.
The first follow-up stopped at design and pseudocode; the second was discussion-only because the round ran out of time.

## PREPARATION

Implement the two-heap median structure and dry-run odd and even stream lengths.
Sketch a bounded-domain alternative and state its update and query costs.
Compare what must change when the requested order statistic is an arbitrary k rather than a fixed median.

**STUDY HERE: topic 4, Heaps, Stacks and Queues**

Also fits: Strings, Parsing and Recursion (listed there as SKIP, no pages). Site tags: string, stack, parsing, medium

MEDIUM | IN APP LC 394 | 1Point3Acres | onsite-coding | 8 of 11

★ TOP 75: rank 60 in the Medium and Hard priority list

# ★ Decode String

ASKED 2026-07-29 | 83 DAYS AGO | LAST 90 DAYS

STUDY: TYPE | ALL TIME 57.7 | 30 DAYS 0 | PRIORITY 45/228

## ABOUT THIS QUESTION

**Topic:** Heaps, Stacks and Queues

**Optimal answer from:** your app's interview approach (LeetCode #394)

**LeetCode match:** LC 394 Decode String

Frequency: Low | Last asked: 2026-07-29 | Roles: SWE, MLE

**Original:** https://www.1point3acres.com/interview/problems/company/google/decode-string

## QUESTION

EXAMPLES

```
"3[a]2[bc]" -> "aaabcbc".
"2[abc]3[cd]ef" -> "abcabccdcdcdef".
"3[a2[c]]" -> "accaccacc".
```

## APPROACH

Write the iterative-stack version from memory in under 10 min. Practice the postfix-syntax variant (parse direction reversed). Drill the decode side together with the encode-shortest dual problem as a pair.

## HOW TO THINK ABOUT IT

Nested repeats like 3[a2[c]] are a bracket structure, which means a stack: when you see '[' save what you have built and the repeat count; when you see ']' repeat the inner string and glue it to the saved prefix. Read counts as full numbers (more than one digit).

**Basis:** LeetCode 394, Decode String.

## BRUTE FORCE

BRUTE FORCE · PSEUDO-CODE

```
def parse():
    result = ""
    while pos < len(s) and s[pos] != "]":
        if s[pos].isdigit():
            n = read_number()
            pos += 1  # skip "["
            inner = parse()
            pos += 1  # skip "]"
            result += inner * n
        else:
            result += s[pos]
            pos += 1
    return result
```

## BRUTE FORCE

| TIME | SPACE |
|---|---|
| O(output length) | O(depth) |

WHY IT IS SLOW

Time is proportional to the output, which is optimal; the cost is the recursion depth, equal to the nesting depth. A stack of (text so far, repeat count) avoids recursion.

## OPTIMAL SOLUTION · Stack of (prefix, repeat) frames

| TIME | SPACE |
|---|---|
| O(output length) | O(output + nesting depth) |

**Code:** see FULL INTERVIEW CODE below (the optimal solution is its main function).

## FOLLOW-UP 1 How would you do it recursively, and what is the risk?

Parse with an index: read letters and digits, on '[' recurse to parse the inner string and return where it ended, on ']' return. It reads naturally but the call depth equals the nesting depth, which can overflow the stack for adversarial inputs. The stack version trades that for explicit memory.

## FOLLOW-UP 2 What if the decoded string is enormous?

Do not materialize it. Return a generator, or compute the character at a given index by walking the structure with lengths (as in LeetCode 880, Decoded String at Index), using index modulo the segment length at each level.

## FOLLOW-UP 3 The decoded string would be enormous.

Compute its length (or a given character) from the stack without building it: keep (length so far, repeat count) frames and combine as prev + inner * times. The length can be computed for strings that could never fit in memory.

## FOLLOW-UP 4 Return the character at position k only.

Walk the structure with lengths: at each repeat reduce k modulo the inner length to find where in the inner string it falls (LeetCode 880 style).

## FOLLOW-UP 5 Invalid input (missing bracket, count with no bracket).

Detect an unmatched ']' (empty stack), a '[' left at the end, and digits not followed by '['. Decide whether to raise or to treat a malformed part as literal text.

## FULL INTERVIEW CODE · base question + every follow-up

**One program for the whole question:** the base answer and every follow-up above, built from small helper functions, with a comment above each step. Type it top to bottom in the interview; the demo at the bottom runs it.

```
# BRUTE FORCE (say it first): Recursive descent over the
# string | time O(output length)
def decode_recursive(s):
    pos = 0

    def parse():
        nonlocal pos
        out = []
        while pos < len(s) and s[pos] != "]":
            if s[pos].isdigit():
                num = 0
                while s[pos].isdigit():
                    num = num * 10 + int(s[pos])
                    pos += 1
                pos += 1  # skip '['
                inner = parse()
                pos += 1  # skip ']'
                out.append(inner * num)
            else:
                out.append(s[pos])
                pos += 1
        return "".join(out)

    return parse()


# BASE QUESTION (optimal): Stack of (prefix, repeat)
# frames
def decode(s):
    # stack of (text so far, repeat count) for each open [
    stack, cur, num = [], [], 0
    for ch in s:
        if ch.isdigit():
            num = num * 10 + int(ch)
        elif ch == "[":
            # remember what we have, start the inner text
            stack.append((cur, num))
            cur, num = [], 0
        elif ch == "]":
            prev, times = stack.pop()
            prev.append("".join(cur) * times)
            cur = prev
        else:
            cur.append(ch)
    return "".join(cur)
```

```
# NESTED FORM (the recursive version)
def parse(s):
    # nested form: items are text or (times, items)
    stack, cur, num = [], [], 0
    for ch in s:
        if ch.isdigit():
            num = num * 10 + int(ch)
        elif ch == "[":
            if num == 0:
                raise ValueError("[ needs a count")
            stack.append((cur, num))
            cur, num = [], 0
        elif ch == "]":
            if not stack:
                raise ValueError("unmatched ]")
            prev, times = stack.pop()
            prev.append((times, cur))
            cur = prev
        else:
            if num:
                raise ValueError("count without [")
            cur.append(ch)
    if stack:
        raise ValueError("missing ]")
    return cur


# FOLLOW-UP 3: The decoded string would be enormous.
def length(items):
    # decoded length without building the string
    total = 0
    for it in items:
        if isinstance(it, str):
            total += 1
        else:
            total += it[0] * length(it[1])
    return total


# FOLLOW-UP 4: Return the character at position k only.
def char_at(items, k):
    # the k-th decoded character (0-based), LeetCode 880
    # style
    for it in items:
        if isinstance(it, str):
            if k == 0:
                return it
            k -= 1
        else:
            times, inner = it
            inner_len = length(inner)
            if k < times * inner_len:
                # which copy does not matter: use k modulo
                return char_at(inner, k % inner_len)
            k -= times * inner_len
    return None


if __name__ == "__main__":
    assert decode("3[a]2[bc]") == "aaabcbc"
    assert decode("2[abc]3[cd]ef") == "abcabccdcdcdef"
    assert decode("3[a2[c]]") == "accaccacc"
    items = parse("3[a2[c]]")
    assert length(items) == 9
    assert [char_at(items, i) for i in range(9)] == list(
        decode("3[a2[c]]"))
    # a huge result is measured, never built
    big = parse("999[999[999[ab]]]")
    assert length(big) == 2 * 999 ** 3
    for bad in ("3[a", "a]", "[a]", "3a"):
        try:
            parse(bad)
            raise AssertionError(bad)
        except ValueError:
            pass
```

## NOTES FROM CANDIDATES

Google variants include alternative syntax (group){k} (postfix repeat), nested grouping with curly braces, and escape characters.

A scale follow-up asks how the interface and memory strategy should change when the encoded string is extremely long. Recursive solution is cleaner but the iterative stack is more robust to bizarre input.

**PREPARATION**

Write the iterative-stack version from memory in under 10 min. Practice the postfix-syntax variant (parse direction reversed). Drill the decode side together with the encode-shortest dual problem as a pair.

**ALSO REPORTED · 1 more**

**String Decompression with Nested Counts | 1Point3Acres**

https://www.1point3acres.com/interview/problems/company/google/string-decompression-nested-counts

Recursion with an index pointer is the cleanest implementation: each call returns the decoded substring of a single (...){k} block plus literals up to its terminator.
An iterative stack-based approach also works: push (StringBuilder, repeatCount) on (, pop and append on ){k}.
The 2-digit k matters only for parsing — read until non-digit.
Watch out for literals between two groups: a(b){2}c(d){3} -> abbcddd.

Preparation: Drill the canonical k[group] decode-string problem to muscle memory; the input grammar there is k[group] instead of (group){k}, but the algorithm is identical.
Practice both recursive and iterative versions — the iterative stack version generalizes more cleanly if the interviewer asks about iterative streaming.
Test with: deeply nested groups, groups containing only literals, two-digit counts ({12}).

**STUDY HERE: topic 4, Heaps, Stacks and Queues**

Also fits: Binary Search and Sorting, Design, Streaming and Scale (listed there as SKIP, no pages). Site tags: dedup, streaming, heap, sorting, medium

MEDIUM | 1Point3Acres | onsite-coding | 9 of 11

# Batch and Streaming Log Deduplication

**ASKED 2026-07-23 | 89 DAYS AGO | LAST 90 DAYS**

**STUDY: READ** | **NO LEETCODE FREQUENCY** | **PRIORITY 226/228**

## ABOUT THIS QUESTION

**Topic:** Heaps, Stacks and Queues

**Optimal answer from:** written and tested here for this exact question

This question has several parts, so each part has its own answer below.

Frequency: Single report | Last asked: 2026-07-23 | Roles: SWE

**Original:** https://www.1point3acres.com/interview/problems/company/google/batch-streaming-log-deduplication

## QUESTION

Process **`(timestamp, message)`** logs by deduplicating on message and sorting retained entries by timestamp, first keeping the first-seen entry and then the latest timestamp. The streaming extension adds **`add(timestamp, message)`** and **`get_next()`**, requiring a map plus min-heap, lazy deletion, and a discussion of watermarks and bounded state.SWEdedupstreamingheapsortingmediumFrequencySingle reportLast asked2026-07-23Stageonsite-coding

## APPROACH

Implement both batch retention policies and write deterministic tie tests.
Build the streaming interface with a membership map, min-heap, and lazy deletion, then state the amortized operation costs.
Rehearse a bounded-memory design using expiry and heap rebuilding, plus a partitioning strategy that keeps each message on one shard.

## INTERVIEW-DAY PLAN

### CLARIFY FIRST

Ask which entry to keep in each policy (first seen or latest timestamp), how ties on timestamp break, whether output is sorted by timestamp, and how the streaming interface returns results (get_next).

### BRUTE FORCE TO SAY OUT LOUD

Sort all logs and dedupe with a set. Correct for the batch version.

### OPTIMAL IDEA

Batch: keep a dict message to chosen entry (keep first-seen, or replace if the new timestamp is later) and sort the values by (timestamp, message). Streaming: a dict for membership plus a min-heap of (timestamp, message); on update of a message push a new entry and mark the old one stale (lazy deletion), and get_next pops until it finds a non-stale one. Amortized O(log n).

### DRY-RUN THESE CASES

The same message twice at different times, equal timestamps, an empty stream, get_next on an empty queue.

### LIKELY FOLLOW-UPS

Watermarks for late data, bounded memory with eviction by age.

### PITFALLS

Returning a stale heap entry. Forgetting deterministic tie-breaking.

## INTERVIEW-DAY PLAN

## HOW TO THINK ABOUT IT

Dedup is a dictionary keyed by the message. Sorting is only for the output order. For a stream, a heap gives the smallest timestamp quickly, but when an entry is replaced you cannot delete from the middle of a heap, so leave the stale entry and skip it when it surfaces (lazy deletion).

**Basis:** Written. Assumed rules (confirm them): dedupe on the message; policy 'first' keeps the earliest in input order, policy 'latest' keeps the largest timestamp; output sorted by (timestamp, message); get_next returns the retained entry with the smallest timestamp and removes it.

## BRUTE FORCE

### BRUTE FORCE · PSEUDO-CODE

```
def brute(logs, keep_first=True):
    kept = []
    for entry in logs:
        same = [x for x in logs
                if x[1] == entry[1]]
        pick = max(same)
        if keep_first:
            pick = same[0]
        if entry == pick:
            kept.append(entry)
    return sorted(kept)  # by timestamp
```

| TIME | SPACE |
|---|---|
| O(n^2) | O(n) |

### WHY IT IS SLOW

Each line scans the others: O(n^2). A dictionary from message to the kept entry is one pass, and a heap with lazy deletion serves the streaming version.

## OPTIMAL SOLUTION · Dictionary on the message, then sort

| TIME | SPACE |
|---|---|
| O(n log n) | O(n) |

**Code:** see FULL INTERVIEW CODE below (the optimal solution is its main function).

## OPTIMAL SOLUTION · Streaming: map + min-heap with lazy deletion

| TIME | SPACE |
|---|---|
| O(log n) amortized per add and get_next | O(n) |

**Code:** see FULL INTERVIEW CODE below (the optimal solution is its main function).

## FOLLOW-UP 1 Late data: how do you decide when it is safe to emit?

Use a watermark: only release entries whose timestamp is at most (the largest timestamp seen minus the allowed lateness). get_next then refuses to return anything newer than the watermark, and anything arriving older than it is dropped or sent to a late-data side output.

## FOLLOW-UP 2 Memory must stay bounded.

Evict messages older than a retention window (a second heap or the same heap keyed on timestamp) and accept that duplicates arriving after eviction are no longer detected. **Cost:** O(log n) per eviction with a heap; memory O(entries inside the retention window).

## FOLLOW-UP 3 Keep the entry with the most occurrences instead.

Store (count, timestamp) per message and update the count on every duplicate; the heap key changes only if you rank by count, so push a new entry when the rank changes and rely on lazy deletion again.

## FOLLOW-UP 4 Only release entries once they are safe (late data may still arrive).

Add a watermark: release only entries whose timestamp is at most (the largest timestamp seen minus the allowed lateness), in order; anything older than the watermark that arrives later goes to a late-data path. The lazy-deletion heap pops stale entries as it goes.

## FULL INTERVIEW CODE · base question + every follow-up

**One program for the whole question:** the base answer and every follow-up above, built from small helper functions, with a comment above each step. Type it top to bottom in the interview; the demo at the bottom runs it.

```
import heapq


# BRUTE FORCE (say it first): Quadratic scan for earlier
# or later duplicates | time O(n^2)
def dedupe_brute(logs, policy="first"):
    keep = []
    for i, (ts, msg) in enumerate(logs):
        if policy == "first":
            ok = all(
                logs[j][1] != msg for j in range(i)
            )
        else:
            ok = all(
                logs[j][1] != msg
                or logs[j][0] < ts
                or (logs[j][0] == ts and j < i)
                for j in range(len(logs))
                if j != i
            )
        if ok:
            keep.append((ts, msg))
    return sorted(keep, key=lambda e: (e[0], e[1]))


# BASE QUESTION (optimal): Dictionary on the message, then
# sort
def dedupe(logs, policy="first"):
    # logs: (timestamp, message) pairs; keep one per
    # message
    kept = {}
    for ts, msg in logs:
        # "first" keeps the first seen, "latest" the
        # newest
        if msg not in kept or (
                policy == "latest" and ts > kept[msg]):
            kept[msg] = ts
    # retained entries sorted by time, then message
    return sorted((ts, msg) for msg, ts in kept.items())


# FOLLOW-UP 4: Only release entries once they are safe
# (late data may still arrive).
class StreamDedup:
    """add(ts, msg) and get_next() with a lazy heap."""
```

```
    def __init__(self, policy="first", lateness=0):
        self.policy, self.lateness = policy, lateness
        # message -> the timestamp currently kept
        self.kept = {}
        # (timestamp, message), may hold outdated entries
        self.heap = []
        self.max_ts = None

    def watermark(self):
        # everything at or below this is safe to release
        if self.max_ts is None:
            return None
        return self.max_ts - self.lateness

    def add(self, ts, msg):
        wm = self.watermark()
        # too late: older than the watermark, drop it
        if wm is not None and ts < wm:
            return False
```

```
        if self.max_ts is None or ts > self.max_ts:
            self.max_ts = ts
        old = self.kept.get(msg)
        if old is None or (
                self.policy == "latest" and ts > old):
            self.kept[msg] = ts
            heapq.heappush(self.heap, (ts, msg))
        return True

    def get_next(self):
        wm = self.watermark()
        while self.heap:
            ts, msg = self.heap[0]
            # lazy deletion: skip entries that were
            # replaced
            if self.kept.get(msg) != ts:
                heapq.heappop(self.heap)
                continue
            # not safe yet: later data may still arrive
            if ts > wm:
                return None
            heapq.heappop(self.heap)
            del self.kept[msg]
            return ts, msg
        return None
```

```
if __name__ == "__main__":
    logs = [(5, "b"), (1, "a"), (9, "a"), (3, "b")]
    assert dedupe(logs, "first") == [(1, "a"), (5, "b")]
    assert dedupe(logs, "latest") == [(5, "b"), (9, "a")]
    s = StreamDedup("first", lateness=2)
    for ts, msg in [(1, "a"), (3, "c"), (5, "b")]:
        assert s.add(ts, msg)
    # watermark is 5 - 2 = 3: release 1, then 3
    assert s.get_next() == (1, "a")
    assert s.get_next() == (3, "c")
    # 5 is still above the watermark
    assert s.get_next() is None
    # data older than the watermark is dropped
    assert not s.add(2, "late")
    # a "latest" policy replaces the kept entry
    t = StreamDedup("latest")
    t.add(1, "x")
    t.add(4, "x")
    assert t.get_next() == (4, "x")
    assert t.get_next() is None
```

## NOTES FROM CANDIDATES

Empty input, identical entries, equal timestamps, out-of-order arrivals, updates, and removals all need explicit semantics.
For keep-latest streaming behavior, stale heap entries must be detected against the current timestamp stored for that message.
Arbitrarily late arrivals make globally final timestamp ordering impossible without a watermark or maximum-lateness guarantee.

## PREPARATION

Implement both batch retention policies and write deterministic tie tests.
Build the streaming interface with a membership map, min-heap, and lazy deletion, then state the amortized operation costs.
Rehearse a bounded-memory design using expiry and heap rebuilding, plus a partitioning strategy that keeps each message on one shard.

**STUDY HERE: topic 4, Heaps, Stacks and Queues**

Also fits: Arrays, Hashing and Two Pointers, Design, Streaming and Scale (listed there as SKIP, no pages). Site tags: heap, hashmap, scaling, medium

MEDIUM | IN APP LC 347 | 1Point3Acres | phone-screen, onsite-coding | 10 of 11

★ TOP 75: rank 29 in the Medium and Hard priority list

# ★ Top-K Frequent in Large Logs

ASKED 2026-07-19 | 93 DAYS AGO | LAST 6 MONTHS

STUDY: TYPE | ALL TIME 58.8 | 30 DAYS 41.1 | PRIORITY 21/228

## ABOUT THIS QUESTION

**Topic:** Heaps, Stacks and Queues

**Optimal answer from:** your app's interview approach (LeetCode #347)

**LeetCode match:** LC 347 Top K Frequent Elements

Frequency: Medium | Last asked: 2026-07-19 | Roles: SWE, MLE

**Original:** https://www.1point3acres.com/interview/problems/company/google/top-k-frequent-from-logs

## QUESTION

EXAMPLES

```
Input:  chatLog = ["A says hi", "B says hi", "A says
    bye", "C says hi", "A says yo"], k=2
Counts: A=3, B=1, C=1
Output: [A, B]  // or [A, C] — ties allowed
```

## APPROACH

Practice writing quickselect with the partition step (and explain how to recover from a bad pivot).
Be able to state amortized vs worst-case time for each approach.
Have a one-line answer for streaming: "hashmap counts + heap of size K maintained online".
Drill the distributed answer once: per-shard top-K, then K-way merge of the partial heaps.

## HOW TO THINK ABOUT IT

Count the speakers first; a log line is just a way to read the user id. Then you only need the K largest counts: a heap of size K does that without sorting all users. For logs spread over many machines the local top K are not enough on their own, so say what extra information you send.

**Basis:** LeetCode 347, Top K Frequent Elements (here the key is the speaker in each chat line).

## BRUTE FORCE

BRUTE FORCE · PSEUDO-CODE

## BRUTE FORCE

```
def brute(chat_log, k):
    names = [l.split()[0] for l in chat_log]
    counts = Counter(names)
    def rank(name):
        return (-counts[name], name)
    return sorted(counts, key=rank)[:k]
```

| TIME | SPACE |
|---|---|
| O(n + d log d) | O(d) |

WHY IT IS SLOW

Sorts all d distinct speakers, O(d log d), when only K are wanted; a size-K heap is O(d log k), and a stream needs a counter sketch instead of exact counts.

## OPTIMAL SOLUTION · Count, then keep a size-K heap

| TIME | SPACE |
|---|---|
| O(n + d log k) | O(d) |

**Code:** see FULL INTERVIEW CODE below (the optimal solution is its main function).

## FOLLOW-UP 1 How do you do it across many machines when the log does not fit on one?

Count per shard and send each shard's local top-K to a coordinator. That is not exact in general (a user can be 11th everywhere yet first overall), so for an exact answer use the threshold algorithm: gather the local top-K, compute a global lower bound, ask every shard for all users above bound / numShards, and merge. If an approximate answer is acceptable, merge count-min sketches (they add) and keep a heap of the K largest estimates.

## FOLLOW-UP 2 What if the log arrives as an unbounded stream?

Use the Misra-Gries or Space-Saving summary with m counters. Any user with more than n / (m + 1) messages is guaranteed to be in the summary, and a second pass (or a count-min sketch for the estimate) can confirm the real counts. **Cost:** O(1) per message with O(m) memory for m counters (Misra-Gries), instead of O(d) memory for the exact counts.

## FOLLOW-UP 3 Which approach would you choose and why?

A heap is the safe default: simple, O(n + d log k) and streaming-friendly. Quickselect wins when k is a large fraction of d and you need all K but not their order. Sorting is fine only when d is small. Say that out loud, with the complexities, because the interviewer is grading the trade-off discussion.

## FOLLOW-UP 4 The log is a stream and memory is limited.

Misra-Gries keeps m counters; any user with more than n / (m + 1) messages is guaranteed to remain in the summary (checked above). A second pass, or a count-min sketch, then gives the real counts of the survivors.

## FULL INTERVIEW CODE · base question + every follow-up

**One program for the whole question:** the base answer and every follow-up above, built from small helper functions, with a comment above each step. Type it top to bottom in the interview; the demo at the bottom runs it.

```
import heapq
from collections import Counter


# BRUTE FORCE (say it first): Count everything, then sort
# all users | time O(n + d log d)
def top_k_speakers_sort(chat_log, k):
    c = Counter(
        line.split(" ", 1)[0] for line in chat_log
    )
    return sorted(c, key=lambda u: (-c[u], u))[:k]


# HELPERS for the base question
def speaker(line):
    # "A says hi" -> "A"
    return line.split(" ", 1)[0]


def rank_key(item):
    # more messages first; ties by name
    return (-item[1], item[0])


# BASE QUESTION (optimal): Count, then keep a size-K heap
def top_k_speakers(chat_log, k):
    counts = Counter(speaker(line) for line in chat_log)
    # a heap of size k: O(n + d log k)
    items = counts.items()
    best = heapq.nsmallest(k, items, key=rank_key)
    return [user for user, _ in best]


# FOLLOW-UP 1: How do you do it across many machines when
# the log does not fit on ...
def merge_shards(shard_logs, k):
    # exact answer: add the full counts of every shard
    total = Counter()
    for log in shard_logs:
        total.update(speaker(line) for line in log)
    best = heapq.nsmallest(k, total.items(), key=rank_key)
    return [user for user, _ in best]


# FOLLOW-UP 2, 4: What if the log arrives as an unbounded
# stream?
class MisraGries:
    """m counters: a user with over n / (m + 1) of the
    n messages always stays in the summary."""

    def __init__(self, m):
        self.m, self.counts = m, {}

    def add(self, user):
        if user in self.counts:
            self.counts[user] += 1
        elif len(self.counts) < self.m:
            self.counts[user] = 1
        else:
            # no room: every counter pays one
            for u in list(self.counts):
                self.counts[u] -= 1
                if self.counts[u] == 0:
                    del self.counts[u]

    def candidates(self):
        return set(self.counts)


if __name__ == "__main__":
    log = ["A says hi", "B says hi", "A says bye",
           "C says hi", "A says yo"]
    assert top_k_speakers(log, 2) == ["A", "B"]
    assert top_k_speakers(log, 5) == ["A", "B", "C"]
    assert top_k_speakers([], 3) == []
    # two shards that each rank someone else first
    s1 = ["A says x", "B says x"]
    s2 = ["B says y", "B says z"]
    assert merge_shards([s1, s2], 1) == ["B"]
    mg = MisraGries(2)
    for u in "AAABBC":
        mg.add(u)
    # A has 3 of 6 messages (> 6 / 3), so it must survive
    assert "A" in mg.candidates()
```

## NOTES FROM CANDIDATES

The interviewer typically wants a discussion of the three implementations + complexity trade-off, not just one working solution.
For the distributed follow-up, candidates that mention count-min sketch and topK per shard -> global merge got positive feedback.
The same family surfaces as "Top-K IPs", "Top-K chatters", and "Top-K words"; the technique is interchangeable.
A recent variant uses a gigabyte-scale IP list and explicitly requires exact time and space complexity analysis.

## PREPARATION

Practice writing quickselect with the partition step (and explain how to recover from a bad pivot).
Be able to state amortized vs worst-case time for each approach.
Have a one-line answer for streaming: "hashmap counts + heap of size K maintained online".
Drill the distributed answer once: per-shard top-K, then K-way merge of the partial heaps.

**STUDY HERE: topic 4, Heaps, Stacks and Queues**

Also fits: Strings, Parsing and Recursion (listed there as SKIP, no pages). Site tags: parsing, recursion, stack, medium

MEDIUM | 1Point3Acres | onsite-coding | 11 of 11

# Nested Function Expression Evaluator

ASKED 2026-06-10 | 132 DAYS AGO | LAST 6 MONTHS

STUDY: TALK | ALL TIME 5 | 30 DAYS 0 | PRIORITY 175/228

## ABOUT THIS QUESTION

**Topic:** Heaps, Stacks and Queues

**Optimal answer from:** written and tested here for this exact question

**LeetCode match:** LC 770 Basic Calculator IV (related)

Frequency: Low | Last asked: 2026-06-10 | Roles: SWE

**Original:** https://www.1point3acres.com/interview/problems/company/google/nested-function-expression-evaluator

## QUESTION

EXAMPLES

```
add(1, sub(1, 0)) -> 2
add(2, mul(3, pow(4, 5))) -> 3074
```

## APPROACH

Write a small tokenizer for identifiers, integers, commas, and parentheses.
Practice a recursive parseExpr() that returns both the value and the next index.
Drill the variable-arity follow-up by collecting arguments until ) instead of assuming exactly two.

## INTERVIEW-DAY PLAN

CLARIFY FIRST

Ask which functions exist, whether they take a fixed or variable number of arguments, whether numbers can be negative, how division behaves, and how errors are reported.

BRUTE FORCE TO SAY OUT LOUD

Repeatedly find the innermost call with a regex and replace it with its value. Quick but fragile.

OPTIMAL IDEA

Tokenize into identifiers, integers, commas and parentheses. Write parse(i) returning (value, next index): a number returns itself; a name must be followed by '(', then loop reading arguments until ')', then apply the function. A function table keeps new operators one line each. O(n), recursion depth equals nesting depth.

DRY-RUN THESE CASES

add(1, sub(1, 0)) is 2. add(2, mul(3, pow(4, 5))) is 3074. A bare number. Variable argument counts. Unknown function names.

LIKELY FOLLOW-UPS

Variables, iterative parsing with two stacks, or division semantics.

## INTERVIEW-DAY PLAN

PITFALLS

Losing the next index after parsing an argument. Treating '-' in '-5' as an operator.

## HOW TO THINK ABOUT IT

An expression with function calls inside function calls has a grammar: expression = number | name '(' expression, expression, ... ')'. Each call can be evaluated only after its arguments, so write a function that parses one expression and reports where it stopped (recursive descent). A tokenizer first, then parse(i) returning (value, next index), keeps the bookkeeping small.

**Basis:** Recursive descent parsing, in the spirit of LeetCode 224 / 726; closest on LeetCode is 770, Basic Calculator IV.

## BRUTE FORCE

BRUTE FORCE · PSEUDO-CODE

```
def evaluate(text):
    while has_call(text):
        call = find_innermost_call(text)
        value = apply(call)  # e.g. add(2, 3)
        text = text.replace(call, str(value))
    return int(text)
```

| TIME | SPACE |
|---|---|
| O(n^2) | O(n) |

WHY IT IS SLOW

Every replacement rescans and rebuilds the string, O(n) per call and O(n^2) in total, and it cannot report errors well. A tokenizer plus recursive descent reads the text once, O(n).

## OPTIMAL SOLUTION · Tokenizer + recursive descent over name(arg, arg, ...)

| TIME | SPACE |
|---|---|
| O(n) | O(depth) |

**Code:** see FULL INTERVIEW CODE below (the optimal solution is its main function).

## FOLLOW-UP 1 Support variadic functions and variables.

Collect arguments until ')' instead of assuming two (as the code does). For variables, pass an environment dict into parse and treat an identifier that is not followed by '(' as a lookup. Define the error behaviour for undefined names.

**FOLLOW-UP 2 How do you make it iterative?**

Use two stacks: one of pending function names and one of lists of evaluated arguments. On an identifier push the name and a new list, on a number append to the top list, on ')' pop both, apply the function, and append the result to the new top list. This removes the recursion-depth limit but is easier to get wrong off by one.

**FOLLOW-UP 3 What about division?**

State the contract: integer or float, truncate toward zero or floor (Python's // floors, which differs for negatives), and divide by zero handling. The implementation above truncates toward zero.

**FOLLOW-UP 4 Add variables (an identifier with no parentheses is a lookup).**

In parse, check whether the identifier is followed by '('; if not, return env[name]. Decide what an undefined name does (raise, or default to 0).

**FULL INTERVIEW CODE · base question + every follow-up**

**One program for the whole question:** the base answer and every follow-up above, built from small helper functions, with a comment above each step. Type it top to bottom in the interview; the demo at the bottom runs it.

```
import math
import re


# BRUTE FORCE (say it first): Repeatedly replace the
# innermost call with its value | time O(n^2)
_OPS = {
    "add": lambda *a: sum(a),
    "sub": lambda a, b: a - b,
    "mul": lambda *a: math.prod(a),
    "pow": lambda a, b: a**b,
    "div": lambda a, b: int(a / b),
    "min": min,
    "max": max,
}


# HELPERS for the base question
TOKEN = re.compile(r"-?\d+|[A-Za-z_]\w*|[(),]")


# NEEDED BY THE BASE QUESTION: it is also the answer to
# the follow-up below
# FOLLOW-UP 3: What about division?
def divide(a, b):
    # integer division that truncates toward zero
    if b == 0:
        raise ZeroDivisionError("division by zero")
    return int(a / b)


OPS = {
    "add": lambda *a: sum(a),
    "sub": lambda a, b: a - b,
    "mul": lambda *a: math.prod(a),
    "pow": lambda a, b: a ** b,
    "div": divide,
    "min": min,
    "max": max,
}


def evaluate_by_replacement(expr):
    pattern = re.compile(r"([A-Za-z_]\w*)\(([^()]*)\)")
    while True:
        m = pattern.search(expr)
        if not m:
            return int(expr.strip())
        args = [int(x) for x in m.group(2).split(",")]
        expr = (
            expr[: m.start()]
            + str(_OPS[m.group(1)](*args))
            + expr[m.end() :]
        )


# BASE QUESTION (optimal): Tokenizer + recursive descent
# over name(arg, arg, ...)
def evaluate(expr, env=None):
    env = env or {}
    tokens = TOKEN.findall(expr)

    def parse(i):
        # returns (value, index after this expression)
        tok = tokens[i]
        if re.fullmatch(r"-?\d+", tok):
            return int(tok), i + 1
        # FOLLOW-UP 4: variables come from env
        # a name not followed by "(" is a variable
        if i + 1 >= len(tokens) or tokens[i + 1] != "(":
            if tok not in env:
                raise ValueError(f"undefined {tok!r}")
            return env[tok], i + 1
        if tok not in OPS:
            raise ValueError(f"unknown function {tok!r}")
        i += 2
        args = []
        # read arguments until the closing bracket
        while tokens[i] != ")":
            value, i = parse(i)
            args.append(value)
            if tokens[i] == ",":
                i += 1
        return OPS[tok](*args), i + 1

    value, end = parse(0)
    if end != len(tokens):
        raise ValueError("trailing tokens")
    return value


# FOLLOW-UP 2: How do you make it iterative?
def evaluate_iterative(expr):
    # two stacks: function names and their argument lists
    names, args, tokens = [], [[]], TOKEN.findall(expr)
    for tok in tokens:
        if re.fullmatch(r"-?\d+", tok):
            args[-1].append(int(tok))
        elif tok == "(":
            # the name just pushed now owns a new list
            args.append([])
        elif tok == ")":
            result = OPS[names.pop()](*args.pop())
            args[-1].append(result)
        elif tok != ",":
            names.append(tok)
    return args[0][0]
```

```python
if __name__ == "__main__":
    assert evaluate("add(1, sub(1, 0))") == 2
    assert evaluate("add(2, mul(3, pow(4, 5)))") == 3074
    # variadic functions
    assert evaluate("add(1, 2, 3, 4)") == 10
    assert evaluate("max(3, min(9, 4), 1)") == 4
    # variables come from an environment
    assert evaluate("add(x, mul(x, 2))", {"x": 5}) == 15
    # division truncates toward zero
    assert evaluate("div(-7, 2)") == -3
    for bad in ("foo(1)", "add(1", "add(1) 2", "y"):
        try:
            evaluate(bad)
            raise AssertionError(bad)
        except (ValueError, IndexError):
            pass
    # the iterative version agrees and has no depth limit
    big = "add(2, mul(3, pow(4, 5)))"
    assert evaluate_iterative(big) == 3074
    deep = "add(1, " * 3000 + "0" + ")" * 3000
    assert evaluate_iterative(deep) == 3000
```

## NOTES FROM CANDIDATES

Tokenizer + recursive descent is the cleanest approach: parse operator, consume (, parse comma-separated child expressions, consume ), then evaluate.
Iterative parsing works but is easy to make off-by-one errors under interview pressure; one candidate was specifically flagged for minor syntax and bounds issues.
Clarify division behavior for negative values and non-divisible integers.

## PREPARATION

Write a small tokenizer for identifiers, integers, commas, and parentheses.
Practice a recursive parseExpr() that returns both the value and the next index.
Drill the variable-arity follow-up by collecting arguments until ) instead of assuming exactly two.